



Computing on Masked Data for Privacy Protection

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Agenda

- Project Overview
- Goals
- Masking Techniques, Threat Models, & Mitigations
- Results
- Applications of Data Masking
- Future Work
- Conclusion



Project Overview

- With privacy preservation becoming an increasingly challenging problem, many models of masking data have been introduced.
- The main goal of this project is to identify approaches and designs for masking data to preserve privacy.



Goals

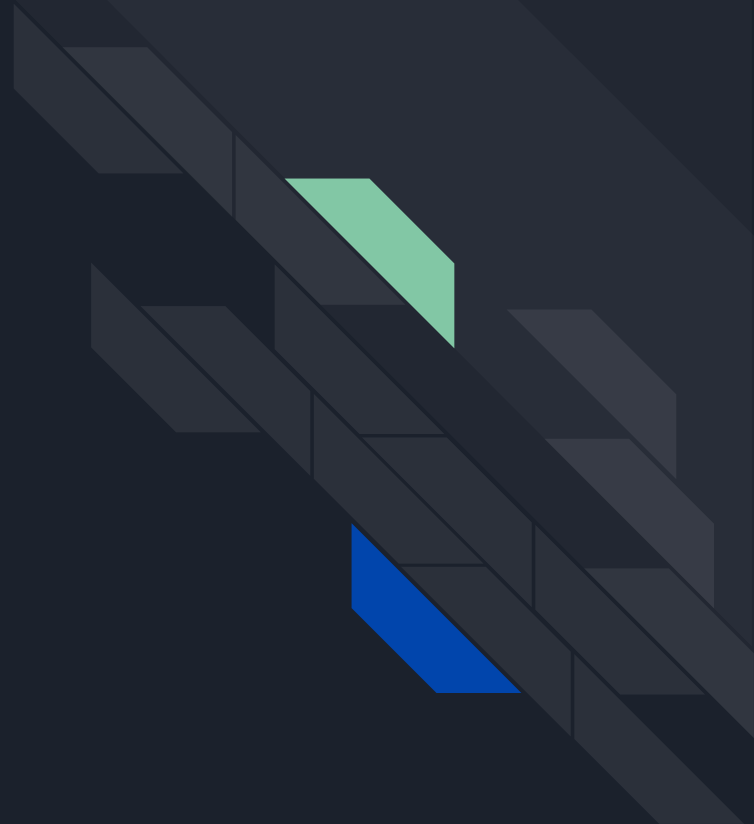
- Analyze existing data masking models
- Survey threats and mitigations for masked data



Guiding Questions

- What advantages and disadvantages are related to the available data masking techniques?
- What is the most efficient AND effective data masking solution?
- Considering known risks and challenges does computing on masked data overcome the risk of leaks due to unmasking?
- How do malicious cyber actors use data masking to defeat cybersecurity?
- Is a hybrid masking architecture feasible?
- Is a hybrid data masking technique feasible?

Data Masking





Overview

- Techniques
 - Encryption
 - Substitution
 - Shuffling
 - Nulling Out
 - Masking Out
- Data utility vs disclosure risk



Encryption

- Uses a key to encode text
- Advantages
 - High cost to attack
 - Low disclosure risk
- Disadvantages
 - High cost to implement
 - Excessive in most cases



Encryption

- Threat
 - Brute force
 - Secret key sharing
- Mitigations
 - Key strength
 - Asymmetric Algorithms



Substitution

- Replace the existing data in the database with random values from a pre-prepared dataset
- Advantages
 - Low disclosure risk
- Disadvantages
 - The pre-prepared dataset must be available
 - Low data utility

Original Data	Masked Data
Aaric	Aaron
Adam	Aaron



Substitution

- Threat
 - Pattern analysis
- Mitigations
 - Use substitutions that are not related to original data
 - Have multiple different possible substitutions



Shuffling

- Swap data fields across different records
- Advantages
 - Good for computing on whole dataset
- Disadvantages
 - High disclosure risk
 - Low data utility when not using whole dataset



Shuffling

- Threat
 - Statistical analysis
- Mitigations:
 - Global sensitivity analysis
 - Laplacian noise



Nulling Out

- Replace the sensitive data with NULL values
 - Phone number -> (XXX)XXX-XXXX
 - Social Security Number (SSN) -> XXX-XX-XXXX
- Advantages
 - No disclosure risk
- Disadvantages
 - No data utility



Masking Out

- Extension of nulling out
- Advantage:
 - Higher data utility
- Disadvantage:
 - High disclosure risk

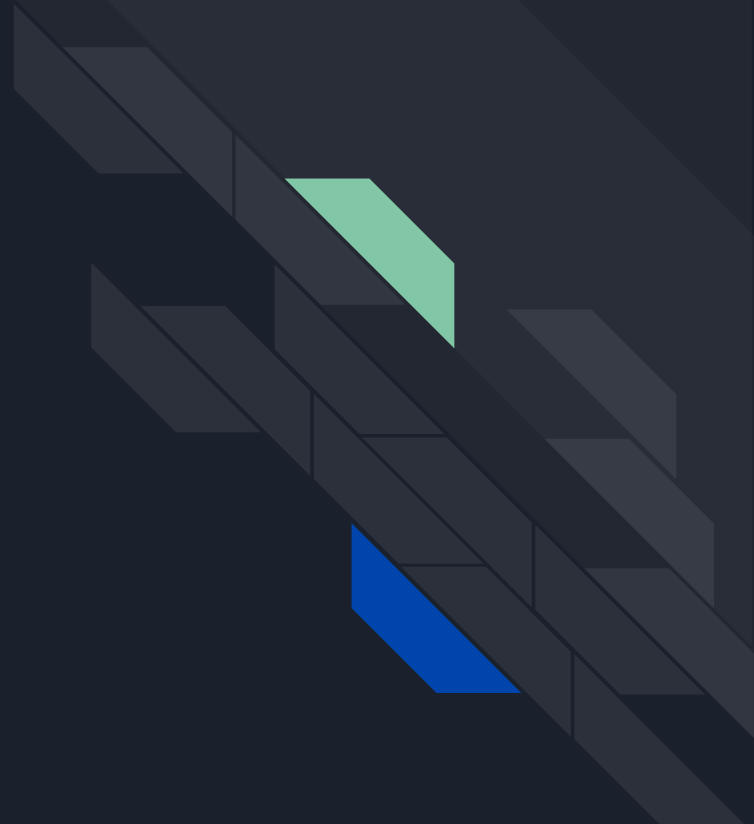
Original	4297 8296 7496 8724
Too Little	4297 8296 7496 87XX
Too Much	XXXX XXXX XXXX XX24
Just Enough	4297 XXXX XXXX 8724



Nulling/Masking Out

- Threat
 - Publicly available information
- Mitigations
 - Masking QIs
 - k-anonymity

Results





Comparison

	Encryption	Shuffling	Substitution	Nulling Out	Masking out
Data type	No	Yes	Yes	No	Partially
Aggregate value	No	Yes	No	No	No*
Frequency distribution	Yes	Yes	No	No*	No*
Uniqueness	Yes	Yes	No	No	No*
Data Disclosure Risk	Low	High	Low	None	High
Data Utility	Low	Low	Low	None	High



Privacy Protection

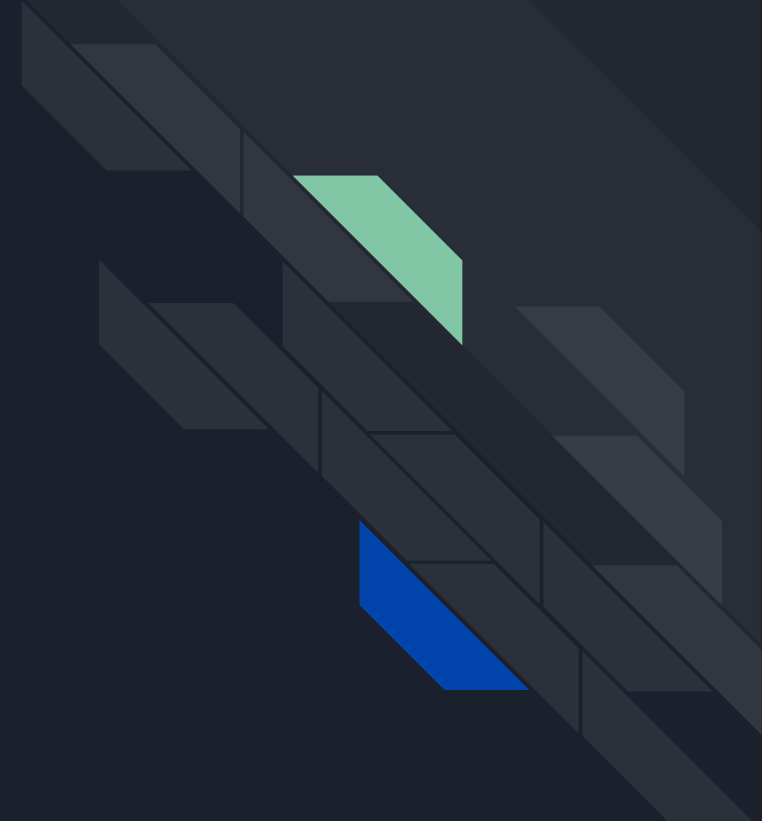
- Partial mitigation of data leaks
- Simply masking data isn't enough
- Varied additional considerations for different techniques
- What would malicious actors want from your dataset?



Hybrid Masking

- Technique
 - Using multiple techniques
 - Strategic combinations of techniques
- Architecture
 - Inferred data

Applications of Data Masking for Cybersecurity Research in Active Defense



Policy-Based Data Masking

- California Consumer Privacy Act (CCPA)
- Healthcare Insurance Portability and Accountability Act (HIPAA)
- General Data Protection Regulation (GDPR)
- The Gramm-Leach-Bliley Act (GLBA)





Source Code Obfuscation

- Prevent code understanding and malicious changes
- Changes structure but keeps functionality
- Types:
 - Layout obfuscation
 - Control flow obfuscation
 - Data obfuscations



Command Obfuscation

- Bypass web application firewalls
- Implicitly run SQL commands against databases
- Example:
 - Converting from string to hex
- Mitigations:
 - Pre processing
 - Prepared statements



Future Work

- Data Masking
 - Working prototype to codify conclusion
 - Is semantic & referential integrity preserved when there are multiple tables in database
 - Dynamic data masking
- Applications
 - Dynamic binary instrumentation
 - Zero-trust framework



Conclusion

- Case-by-case algorithms
- Scope of research

Questions

